Prototype Discriminative Learning for Semi-Supervised Change Detection in Remote Sensing Images

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***Abstract*— With the continuous progress of deep learning in remote sensing (RS) visual tasks, considerable advancements have been achieved in RS image change detection (CD). However, prevailing CD methods heavily rely on extensive sets of fully pixelwise hand-annotated training data, a time-consuming and costly process, and they fail to fully harness the potential benefits of deep feature representations within the deep feature domain. To tackle the mentioned issues, we propose a novel semi-supervised CD method called PDLCD, which strategically leverages useful information from massive unlabeled data to complement labeled data with just a few samples. Specifically, changed objects and unchanged backgrounds of bitemporal RS images are various and complex, our approach advocates dividing each category into multiple subclasses in the deep feature domain. In this scheme, the high-level feature of each subclass follows a Gaussian distribution. Then, the prototype discriminative learning (PDL) is introduced to explicitly encourage deep features of samples closer to the nearest prototype within their respective category, and away from all prototypes of other categories. We design feature discriminative loss (FDL) to implement PDL for constructing more pronounced intraclass compactness and interclass variability. Finally, we compute the supervised loss based on a limited set of labeled data, incorporate the unsupervised loss leveraging a substantial volume of unlabeled data, and include FDL within the deep feature domain to collectively optimize the model. Extensive experiments carried out on three challenging RS image CD datasets illustrate that our proposed semi-supervised CD method obtains better CD performance than previous counterparts. The source code is available at: https://github.com/Youzhihui/PDLCD.**

***Index Terms*— Change detection (CD), deep learning, prototype, remote sensing (RS), semi-supervised learning.**

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I. Introduction

AT PRESENT, with the continuous development and advancement of remote sensing (RS) technology, a large amount of high-resolution RS data are easily accessible and processed, these data provide excellent opportunities to monitor and assess the changes in surface of the Earth. In the realm of RS image processing, change detection (CD) has become one of the most attractive tasks. The goal of CD is to discern changed information, including surface cover type, structure, shape, and location. This is achieved by densely assigning either unchanged (0) or changed (1) labels to each pixel in bitemporal or multitemporal images captured at different periods of the same geographical region. CD of RS image is widely applied in the domains of urbanization monitoring [1], land cover surveys [2], and natural disaster evaluation [3].

As deep learning has undergone rapid development in the past decade, traditional supervised CD methods [4], [5] have witnessed remarkable success. These works design reasonable and effective end-to-end models tailored to the unique characteristics of CD task. Some single-stream approaches [6] concatenate original RS bitemporal images into a whole input for segmentation networks, but they fail to fully exploit the helpful deep features present in each individual raw image. In contrast, more supervised CD methods adapt double-stream models [5], [7], [8], employing Siamese networks such as convolutional neural network (CNN) and transformer structure as encoders for handling RS bitemporal images separately. To further expand receptive field, deeper CNN backbones [9] or dilated convolution [10] are utilized to extract more robust changed features. Some methods [4], [11] model long-range dependencies by using transformer to capture global contexts of changed areas. Moreover, other methods [12] integrate the advantages of CNN and transformers to further improve representative features, facilitating effective and comprehensive detection of changed objects. However, these fully supervised CD methods suffer from the urgent drawback of heavily relying on large amounts of labeled training data. In real application scenarios, the manual annotation of extensive labeled data is a laborious and expensive undertaking.

To mitigate the reliance on extensive labeled data, researchers have delved into semi-supervised learning algorithms. Initial semi-supervised approaches are applied to image classification task by using consistency regularization [13],

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[14] and entropy minimization [15], [16] paradigms. Recently, more semi-supervised semantic segmentation algorithms [17], [18], [19] have been proposed to tackle the challenge of high annotation costs for pixelwise data. These approaches have demonstrated excellent performance even with limited labeled training samples. As a result, some works [20], [21], [22] have been made to involve semi-supervised learning algorithms in CD task, which could be broadly categorized into adversarial-learning-based (ALB) and consistency-learning-based (CLB) approaches. ALB semi-supervised CD methods, such as SemiCDNet [23] and GDCN [24], utilize adversarial neural networks to enable models to discriminate between real and fake changed areas. However, they are often susceptible to unstable adversarial optimization strategies. Therefore, more CLB semi-supervised CD methods [25] have emerged to improve performance by adapting consistency regularization and pseudolabel. As for the scheme of pseudolabel, some methods [20], [21] filter out the low-confidence pixels by different strategies, and then combine reliable pseudolabel to improve performance and generalization of models. In contrast, Wang et al. [26] divided reliable and unreliable pixels based on the predicted entropy, introducing contrastive loss to fully explore those unreliable data. These semi-supervised CD methods hold promise in case of limited labeled data.

However, existing semi-supervised CD methods often leverage the decision boundary of classifiers, neglecting critical representative information embedded in the deep feature domain. In classification tasks, metric learning [27] and contrastive learning [28] are frequently employed to improve the representation capability by minimizing the differentiation between similar samples and maximizing it between samples from distinct categories. Therefore, based on the distance metric, prototype discriminative learning (PDL) is proposed in this work to further improve the CD performance. Unlike conventional contrastive learning, PDL avoids seeking positive and negative samples. Due to the variety of changed objects and the complexity of background in binary CD task, it assumes that each category comprises multiple subclasses, ensuring that deep features within each subclass adhere to a Gaussian distribution in the deep domain. Samples are guided toward the center (expectation or prototype) of the most similar subclass within the category to which they belong, while staying away from all prototypes of other categories. This objective is achieved through feature discriminative loss (FDL).

In this article, a novel semi-supervised CD approach based on the popular student–teacher (ST) [16] framework, called PDLCD, is proposed to explore the discriminability of deep features by using labeled and unlabeled data. Almost all semi-supervised methods of ST variants add weak and strong perturbations on unlabeled data to guide the unsupervised training. They employ a moderate threshold on prediction probability or entropy to select reliable pseudolabel, extracting useful information from unlabeled data. However, they often ignore crucial cues present in the deep feature domain, which can be harnessed directly without the need for learnable classifiers. Therefore, we propose PDL at the high-level deep feature extracted by the encoder. In the binary CD task, changed and unchanged objects exhibit variability and complexity. We assume that each category can be divided into multiple subclasses without affecting their diversity (e.g., changed objects can be further divided into changed buildings, roads, vegetation, or others, while unchanged areas can also be grouped into different backgrounds, including unchanged fields, forests, or others). We design FDL in the deep feature domain, which can prompt that samples are pulled toward the nearest prototype and pushed away from all prototypes of other categories. PDL effectively establishes intraclass compactness and interclass variability, helping to improve the CD performance.

The main contributions of this work are as follows.

1) A novel semi-supervised CD method based on PDL is proposed, which can obtain better performance by combining a few labeled data and massive unlabeled data.
2) FDL is designed in the deep feature domain, which further establishes intraclass compactness and interclass variability to promote the distinctiveness of different classes, improving the performance of detecting changed objects.
3) Experimental results on three public CD datasets illustrate that PDLCD achieves better performance compared with other methods, further validating its effectiveness.

The rest of this article is organized as follows. Section II discusses related semi-supervised methods and CD works in detail. Our proposed semi-supervised CD method is described in Section III. Section IV presents the experimental results, while Section V provides further debate. Finally, Section VI summarizes the conclusion of this work.

II. Related Works

In this section, we begin with an introduction to recent fully supervised CD methods. Next, semi-supervised semantic segmentation is briefly described, followed by a description of semi-supervised CD approaches.

A. Fully Supervised CD Approaches

Fully supervised deep learning CD methods can be categorized into single-stream and double-stream based on different fusion phases. In general, single-stream methods [29], [30] concatenate original RS bitemporal image pairs to match the single input of segmentation networks, but they fail to extract deep features of a single image to complement the detection process. In contrast, double-stream methods [10], [31], [32] generally employ Siamese backbones as feature extractors, processing bitemporal images independently. These double-stream approaches usually exhibit superior performance due to the feature fusion after bitemporal features are extracted separately. On the other hand, based on different network structures, existing supervised CD methods can also be grouped into CNN-based [5], [32], transformer-based [4], or hybrid structure [12], [31]. A series of dense skip connections [5], [33] or multilevel feature aggregations [34] have been designed to compensate for missing information about changed objects. Moreover, deeper CNN backbones [9] or

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dilated convolution [10] are used to build richer semantics about changed areas, but they hold more parameters and higher computational costs. These CNN-based CD methods have limited efficiency in capturing global information, making poor performance when handling long-range dependencies in RS images. Transformer architecture [11], [35] has become popular in recent years for CD task due to its ability of modeling global information. For example, Chen et al. [4] described changed regions with some high-level semantic tokens and built contexts of compact token-based space-time by transformer encoder, but additional redundant contexts and computational complexity are introduced. In addition, some hybrid structure methods [12], [31] have been developed to simultaneously extract local information and model global content for better suiting the CD task. Although these methods mentioned above have showcased outstanding CD performance in complex scenarios, they require a substantial volume of fully hand-annotated training data at the pixel level, and the annotation process is an extremely laborious and costly operation.

B. Semi-Supervised Semantic Segmentation

Conventional CD methods usually employ semantic segmentation paradigm based on the pixelwise classification task. Semi-supervised learning algorithms [16], [26] can effectively alleviate the reliance on extensive annotated data. Numerous emerging semi-supervised semantic segmentation algorithms [18], [19], [26], [36] have been presented with great success. To fully leverage consistency regularization, one of the basic semi-supervisory paradigms, CCT [17] added different perturbations to intermediate features for consistent output between the main decoder and several auxiliary decoders. These consistent regularization semi-supervised methods [17], [19], [26] with multiple forward propagations increase computational costs, and they are affected by the distribution of input data. As for pseudolabel semi-supervised paradigms, Chen et al. [37] proposed cross-pseudosupervision (CPS), utilizing pseudolabel generated by one model to supervise another model with the same structure. To implement a more reliable and accurate pseudolabel, Yang et al. [18] proposed an advanced semi-supervised framework ST++. However, many unlabeled data are underutilized due to their unreliability on [19] and [36]. In response to this issue, Wang et al. [26] introduced the contrastive loss to fully exploit the potential of unreliable pseudolabel. As a result, these semi-supervised methods have achieved excellent performance with limited hand-annotated data and have inspired the emergence of semi-supervised CD approaches.

C. Semi-Supervised CD Approaches

With the popularity of semi-supervised learning, there has been a growing focus on semi-supervised CD approaches, which could be roughly grouped into two types: ALB and CLB. In the context of ALB semi-supervised CD methods, Gong et al. [24] developed a CD model for multispectral images based on a generative discriminatory classified network, using labeled, unlabeled, and fake data to train the CD model. To fully exploit massive unlabeled data, SemiCD-Net [23] was presented based on a generative adversarial network, it employed two discriminators to ensure the consistency of feature distribution and minimized two adversarial losses. However, these ALB methods usually suffer from unstable adversarial optimization strategies, and CD performance becomes poor when annotated samples are limited. Therefore, more CLB methods [22], [25], [38], [39] have been presented recently. Bandara and Patel [25] proposed RCR based on the consistency regularization, they added different random perturbations to differential features constraining the consistency of outputs from different decoders, with the disadvantage of more parameters and computational costs. To enhance the variability of deep features between different categories, Wang et al. [20] designed simple and effective contrastive loss based on the changed regions to improve CD performance. In addition, FPA [21] learned area-invariant and augmentation-invariant feature representation from unlabeled RS images by two alignment strategies. Although these CLB methods have demonstrated impressive performance with limited annotated data, they primarily devote to improving the discrimination of the decision boundary of the classifier, while ignoring critical cues embedded in the deep feature domain. Therefore, PDL is introduced to enhance the discriminability of deep features, effectively promoting intraclass compactness and interclass variability, which helps improve the CD performance.

III. Proposed Approach

A. Overview

In semi-supervised CD task, we are given a smaller set of fully pixelwise annotated training images $\mathcal{D}^l = \{(x_A^i, x_B^i, y_i)\}_{i=1}^{N_l}$ and a much larger unlabeled images $\mathcal{D}^u = \{(u_A^i, u_B^i)\}_{i=1}^{N_u}$, where A and B denote different time periods corresponding to the acquisition of bitemporal images, N_l and N_u indicate the number of labeled images and unlabeled images, and $N_l \ll N_u$ in most cases. x^i or $u^i \in \mathbb{R}^{H\times W\times D}$ means input images with a size of $H \times W$ and D channels, $y_i \in \{0,1\}^{H\times W\times C}$ is the one-hot ground-truth label for all pixels corresponding to (x_A^i, x_B^i), where $C = 2$ indicates the number of categories (changed and unchanged) in the CD task. Our ultimate goal is to train a CD model with the excellent performance by effectively leveraging limited labeled and numerous unlabeled bitemporal images.

The overall structure of our proposed approach is depicted in Fig. 1, our method is built on the popular ST framework for semi-supervised learning. Specifically, the student model and teacher model share an identical network structure but undergo distinct updates during the training process. These parameters of student model, denoted by θ_s, are updated based on the conventional loss gradient, while the parameters of teacher model, denoted by θ_t, are updated by performing exponential moving average (EMA) with parameters of the student model. The primary responsibility of teacher model is to furnish a reliable pseudolabel for unlabeled images,

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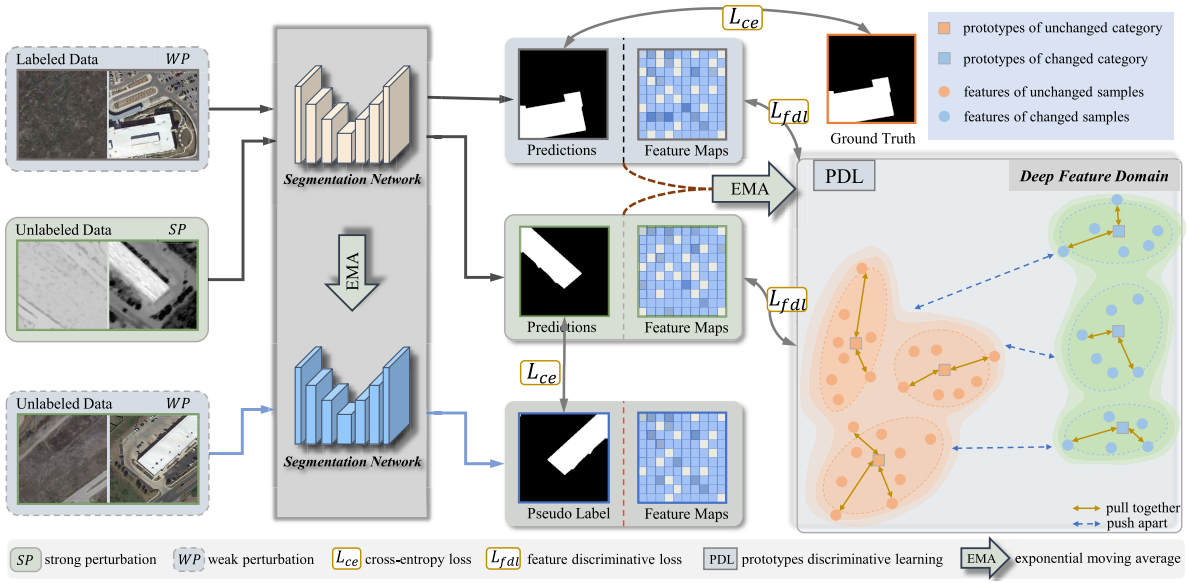


Fig. 1. Overview of the proposed semi-supervised CD method based on PDL. Our method is build upon the popular ST frameworks. The FDL is designed to encourage features of samples closer to the nearest prototype while away from all prototypes of other categories in the deep feature domain. These prototypes are updated using feature maps of labeled and unlabeled data for EMA. The illustration of FDL is presented in Fig. 2.

enabling the student model to adequately leverage information from a substantial pool of unlabeled data as a supplement, improving the generalization and robustness of the model.

The input of our framework consists of three data in one forward pass: one labeled data input (x_A, x_B) with random weak perturbation, the weakly perturbed view (u_A, u_B) and strongly perturbed view $(\widetilde{u}_A, \widetilde{u}_B)$ transformed from unlabeled data. For each labeled image, our aim is to minimize the standard cross-entropy loss with ground truth, and the formulation of supervised loss is articulated as follows.

$$L_s = \frac{1}{|\mathcal{D}^l|} \sum_{(x_A^i, x_B^i, y_i) \in \mathcal{D}^l} L_{\text{ce}}\left(f_s\left(x_A^i, x_B^i\right), y_i\right) \tag{1}$$

where f_s is the student model and $L_{\text{ce}}(\cdot)$ denotes the cross-entropy loss function. As for each unlabeled image, we first take weakly perturbed view (u_A, u_B) into the teacher model and get prediction to generate a pseudolabel, which can be expressed as follows:

$$\hat{y} = \text{argmax}\, f_t(u_A, u_B). \tag{2}$$

As is customary in many semi-supervised approaches, we then introduce a threshold γ ignoring unreliable pseudolabel and retaining only reliable ones for optimizing student model. When it comes to calculating unsupervised loss, the strongly perturbed view $(\widetilde{u}_A, \widetilde{u}_B)$ is entered into student model

$$L_u = \frac{1}{|\mathcal{D}^u|} \sum_{(u_A^i, u_B^i) \in \mathcal{D}^u} L_{\text{ce}}\left(f_s\left(\widetilde{u}_A^i, \widetilde{u}_B^i\right), \hat{y}_i\right). \tag{3}$$

At the same time, the scarcity of labeled data often proves inadequate for establishing both intraclass compactness and interclass variability. To alleviate this problem, we propose PDL. Specifically, diverging from prior semi-supervised approaches that solely introduce additional unsupervised losses, FDL is incorporated into both labeled and unlabeled data. PDL is motivated by the popular metric learning principle, which involves maximizing the distance (minimizing the similarity) between interclass pixels and minimizing the distance (maximizing the similarity) between intraclass pixels within the deep feature domain. A detailed exposition of this principle will be provided in Section III-B. To summarize, the overall loss is calculated as follows:

$$L_{\text{all}} = L_s + \lambda_1 \cdot L_u + \lambda_2 \cdot L_{\text{fdl}} \tag{4}$$

where L_{fdl} denotes the FDL to be explained in detail in Section III-B and λ_1 and λ_2 are the hyperparameters to balance total loss.

B. Prototype Discriminative Learning

For superior CD models, the downstream linear classifier involved can assign learnable parameters to each category. Concurrently, the encoder plays a crucial role in proficiently extracting and leveraging promising deep feature representations. This implies that samples from the same category ought to be closely associated with the nearest prototype within that category while maintaining a significant distance from all prototypes of other categories within the deep feature domain. In other words, this setup encourages intraclass samples to exhibit proximity, promoting intraclass compactness, while simultaneously pushing interclass samples as far apart as possible to enhance interclass variability.

Here, it is necessary to introduce specific symbols that will facilitate subsequent discussions. Precisely, we denote x_{ij} and u_{ij} as the input data for the ith labeled image at pixel j and the ith unlabeled image at pixel j, respectively. The softmax outputs from the segmentation head of student model corresponding to these inputs are denoted by q_{ij}^l and q_{ij}^u. Furthermore, the intermediate deep features of x_{ij} and u_{ij} are denoted by $\boldsymbol{r}_{ij}^l$ and $\boldsymbol{r}_{ij}^u$. The ground truth of labeled input

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is marked as y_{ij}, while $\hat{y}_{ij}$ indicates the pseudolabel produced by the teacher model for the unlabeled data.

The prototype, serving as either a sample feature or the average of a set of sample features from the same class, offers flexibility in representation. For binary CD tasks, changed objects are various and backgrounds are complex, the concept of a category extends to multiple prototypes to preserve diversity effectively. Without loss of generality, we denote the prototype set as $\mathcal{P} = \{\boldsymbol{\mu}_{m,n} | 1 \le m \le C, 1 \le n \le T\}$, where C denotes the number of categories in the CD task (changed or unchanged), and T means the number of prototypes for each category (the numbers can be different). Each prototype possesses a covariance, expressed as $\mathcal{C} = \{\boldsymbol{\Sigma}_{m,n} | 1 \le m \le C, 1 \le n \le T\}$. Our assumption is that the deep features of each subclass belonging to a prototype obey a Gaussian distribution, and then, each category consists of multiple Gaussian models to form a Gaussian mixture model (GMM). Considering that the downstream deep feature of input data is high-dimensional, the probability density function of the Gaussian distribution is

$$p(x|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = N(x|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{D}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp\left(-\frac{(x-\boldsymbol{\mu})^{\mathrm{T}} \boldsymbol{\Sigma}^{-1} (x-\boldsymbol{\mu})}{2}\right) \quad (5)$$

where D is the data dimension, $\boldsymbol{\mu}$ is the data mean (expectation), $\boldsymbol{\Sigma}$ indicates the covariance, and T denotes matrix transpose. GMM consisting of category c can be expressed as follows:

$$\Phi_c(x|m=c) = \sum_{n=1}^{T} \omega_{c,n} \cdot p(x|\boldsymbol{\mu}_{c,n}, \boldsymbol{\Sigma}_{c,n}). \quad (6)$$

In the deep feature domain, a fundamental observation is that deep features belonging to the same category should exhibit proximity to the nearest prototype within that category, while concurrently maintaining a considerable distance from all prototypes belonging to other categories. This concept is visually illustrated in Fig. 2. Therefore, the probability score indicating the affiliation of a particular sample with the tth prototype in category c can be expressed as follows:

$$s_{ij}(c,t) = \begin{cases} s^l_{ij}(c,t), & \text{if image } i \text{ is labeled} \\ s^u_{ij}(c,t), & \text{otherwise} \end{cases} \quad (7)$$

where $s^l_{ij}(c,t)$ and $s^u_{ij}(c,t)$ correspond to labeled and unlabeled data, and they are calculated by (8) and (9), respectively.

For ith labeled image, calculating the score of jth pixel with each prototype in category c should be: 1) the ground truth of this sample belongs to c and 2) each score with one prototype in proportion to the total score of all prototypes

$$s^l_{ij}(c,t) = \frac{p(\boldsymbol{r}^l_{ij}|\boldsymbol{\mu}_{c,t}, \boldsymbol{\Sigma}_{c,t}) \cdot I[y_{ij} = c]}{\sum_{m=1}^{C} \sum_{n=1}^{T} p(\boldsymbol{r}^l_{ij}|\boldsymbol{\mu}_{m,n}, \boldsymbol{\Sigma}_{m,n}) \cdot I[y_{ij} = c]} \quad (8)$$

where $I(\cdot)$ is the indicator function. For ith unlabeled image at jth pixel $s^u_{ij}(c,t)$, it is computed similar to the labeled image, with the difference that reliable pseudolabel generated by teacher model are leveraged to determine the category

$$s^u_{ij}(c,t) = \frac{p(\boldsymbol{r}^u_{ij}|\boldsymbol{\mu}_{c,t}, \boldsymbol{\Sigma}_{c,t}) \cdot I[\hat{y}_{ij} = c]}{\sum_{m=1}^{C} \sum_{n=1}^{T} p(\boldsymbol{r}^u_{ij}|\boldsymbol{\mu}_{m,n}, \boldsymbol{\Sigma}_{m,n}) \cdot I[\hat{y}_{ij} = c]}. \quad (9)$$

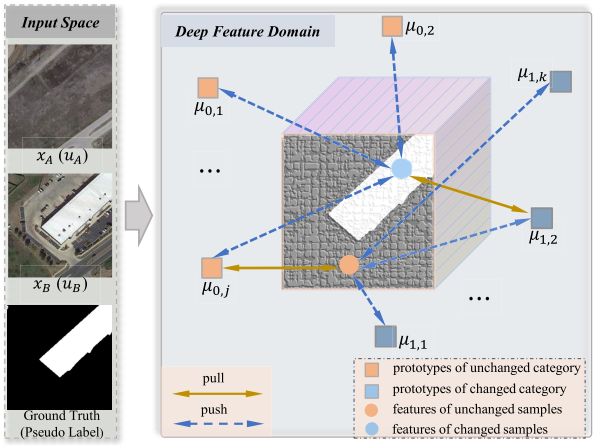


Fig. 2. Schematic of FDL. The features of samples are pulled toward the nearest prototype while being pushed away from all prototypes of other categories. $\mu_{m,n}$ denotes the nth prototype of the mth category.

Next, we identify the nearest prototype and mark it via

$$(v, w) = \underset{m,n}{\operatorname{argmax}}\, s_{ij} \quad (10)$$

where (v, w) means that the most similar prototype label corresponds to the wth prototype within category v.

Finally, our objective is to move closer to the nearest prototype and simultaneously move away from prototypes belonging to other categories. In terms of probability, the Gaussian distribution of the nearest prototype results in a higher probability score, while the scores for other diverse categories are comparatively diminished. Therefore, minimizing the following prototype-based FDL for labeled data can achieve it

$$L^l_{\mathrm{fdl},ij} = 1 - p\left(\boldsymbol{r}^l_{ij}|\boldsymbol{\mu}_{v,w}, \boldsymbol{\Sigma}_{v,w}\right) + \sum_{m \ne v} \sum_{n=1}^{T} \omega_{m,n} \cdot p\left(\boldsymbol{r}^l_{ij}|\boldsymbol{\mu}_{m,n}, \boldsymbol{\Sigma}_{m,n}\right) \quad (11)$$

where values of the second and third terms both lie between 0 and 1. Note that coefficient 1 is to ensure that the value of loss is always nonnegative. For unlabeled data, counterpart $L^u_{\mathrm{fdl},ij}$ is calculated in the same way using (11). The two losses are combined and described as follows:

$$L_{\mathrm{fdl}} = \frac{1}{|\mathcal{D}^l|} \sum_{(x^i_A, x^i_B, y_i) \in \mathcal{D}^l} \sum_{j} L^l_{\mathrm{fdl},ij} + \frac{1}{|\mathcal{D}^u|} \sum_{(u^i_A, u^i_B) \in \mathcal{D}^u} \sum_{j} L^u_{\mathrm{fdl},ij} \cdot I\left[q^u_{ij} > \gamma\right] \quad (12)$$

where γ denotes the threshold, ensuring that only reliable pseudolabel actively contribute to the gradient update for unlabeled data. Fig. 3 presents the distribution process of prototypes and samples of different categories in the deep feature domain during training.

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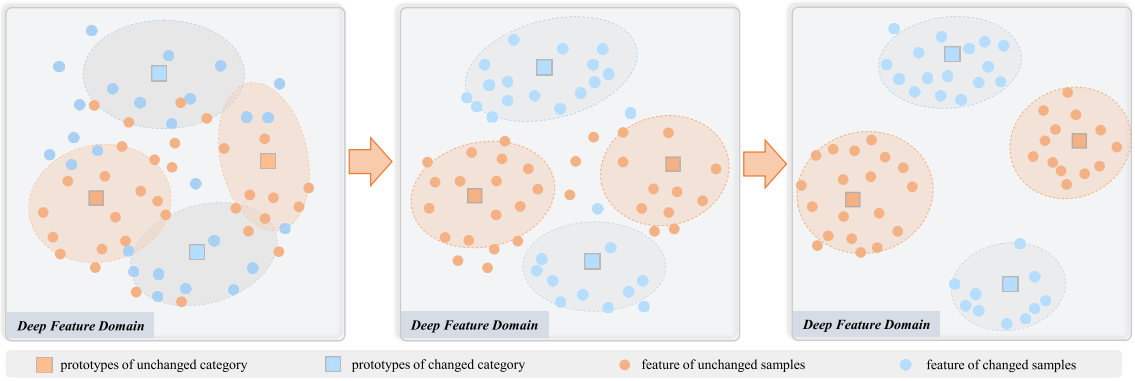


Fig. 3. Illustration of the distribution process of prototypes and samples of different categories in the deep feature domain during the model is trained.

Algorithm 1 Prototype and Covariance Initialization

Input:
Labeled data set $\mathcal{D}^l = \{(x_A^i, x_B^i, y_i)\}_{i=1}^{N_l}$;
Number of prototypes T;
Output:
Initial prototypes and covariance;
Train student-model $f_s(\theta)$ on $\mathcal{D}^l$ with L_s;
$\mathcal{S}_0 \leftarrow \emptyset$, $\mathcal{S}_1 \leftarrow \emptyset$;
for (x_A^i, x_B^i, y_i) in $\mathcal{D}^l$ **do**
  $\boldsymbol{r}_i^l, y_i^p = f_s(x_A^i, x_B^i)$;
  Add deep feature representations $\boldsymbol{r}_i^{l_0}$ of unchanged samples corresponding to $y_i^p == y_i$ and y_i is unchanged category to $\mathcal{S}_0$;
  Add deep feature representations $\boldsymbol{r}_i^{l_1}$ of changed samples corresponding to $y_i^p == y_i$ and y_i is changed category to $\mathcal{S}_1$;
end for
Use deep feature set $\mathcal{S}_0$ of the unchanged samples to obtain shared covariance for unchanged category;
Use deep feature set $\mathcal{S}_1$ of the changed samples to obtain shared covariance for changed category;
Create T sub-classes for unchanged category and T sub-classes for changed category performing K-Means clustering (Mahalanobis distance is used) on sampled pixel representations from $\mathcal{S}_0$ and $\mathcal{S}_1$, respectively;
Calculate the feature average of samples in each sub-class to obtain the initial prototypes of each category;
Calculate the feature covariance of samples in each sub-class to obtain the initial covariance of each category.

C. Prototype and Covariance Initialization and Update

1) Prototype and Covariance Initialization: To initialize the prototype, we first train the CD network in a fully supervised manner (adopting student model with gradient update) using the given labeled images $\mathcal{D}^l$ with several epochs. Then, we utilize the trained student model to extract pixelwise feature representations for each category (change and unchanged) within the given labeled training data. To be specific, when the predicted result y_{ij}^p is consistent with the given ground truth y_{ij}, we incorporate the corresponding feature representation $\boldsymbol{r}_{ij}^l$ into the representation set of the respective category. With the set of intraclass pixelwise feature representations in hand, we proceed to compute the feature covariance for each category. Following this, we perform K-means clustering based on the Mahalanobis distance measure to divide multiple internal subclasses. Note that the number of subclasses (or prototypes) for each category is predefined. Next, the initial prototypes are computed by averaging feature representations within the same subclass, simultaneously determining the initial covariance matrices for each subclass. For a detailed of the prototype and covariance initialization process, please refer to Algorithm 1.

2) Prototype Update: In our approach, prototypes are dynamically updated from downstream deep features extracted in labeled images and unlabeled samples during training. We assign each pixel in current mini-batch to one of the prototypes based on two conditions: 1) the assigned prototype should belong to the same category as the pixel and 2) the assigned prototype should exhibit the highest similarity among all other prototypes in the same category. We denote the feature set of all labeled candidate pixels belonging to the tth prototype (subclass) in category c as $\mathcal{R}_{c,t}^l$, which is expressed by the following equation:

$$\mathcal{R}_{c,t}^l = \left\{ \boldsymbol{r}_{ij}^l | y_{ij} = c, q_{ij}^l(c) > \gamma, t = \operatorname{argmax} s_{ij,c}^l(m) \right\} \quad (13)$$

where y_{ij} is ground truth for the jth pixel in labeled image i, γ denotes the threshold, $s_{ij,c}^l$ [referencing (8)] signifies the scores of this sample with all prototypes in category c, and $\boldsymbol{r}_{ij}^l$ indicates representation of the jth pixel in labeled image i. For unlabeled data, counterpart $\mathcal{R}_{c,t}^u$ can be calculated in a similar fashion using (13). Also, the difference between them is that the former relies on hand-annotated label, while the latter utilizes pseudolabel $\hat{y}_{ij}$ derived from the teacher model. Therefore, for the tth prototype of category c, the set of qualified features belonging to that prototype is expressed as follows:

$$\mathcal{R}_{c,t} = \mathcal{R}_{c,t}^l \cup \mathcal{R}_{c,t}^u. \quad (14)$$

The prototypes are dynamically updated within each mini-batch, so we need to compute the average features of the represents set belonging to a prototype in

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the current mini-batch

$$\boldsymbol{\mu}_{c,t}^{\text{cur}} = \frac{1}{|\mathcal{R}_{c,t}^{\text{cur}}|} \sum_{\boldsymbol{r}_{c,t} \in \mathcal{R}_{c,t}^{\text{cur}}} \boldsymbol{r}_{c,t}. \tag{15}$$

Once the assignment is done, we similarly apply the EMA strategy to update the prototype using the following formula:

$$\boldsymbol{\mu}_{c,t}^{\text{new}} = \alpha \cdot \boldsymbol{\mu}_{c,t}^{\text{old}} + (1-\alpha) \cdot \boldsymbol{\mu}_{c,t}^{\text{cur}} \tag{16}$$

where α is a hyperparameter controlling the update speed.

3) Covariance Update: Each set of features for a subclass corresponds to a covariance. Unlike the prototype update mechanism, the covariance is not updated within the mini-batch, rather it is adjusted after each epoch. However, the model is trained in a mini-batch, and it is impractical to store all sample features to calculate the covariance when an epoch is over. Therefore, we dynamically update the covariance online between different mini-batches in an epoch. For the current mini-batch, the current average feature $\mu_{c,t}^{\text{cur}}$ of the tth subclass in category c is obtained by (16), and the corresponding current feature covariance can be expressed as follows:

$$\boldsymbol{\Sigma}_{c,t}^{\text{cur}} = \frac{1}{|\mathcal{R}_{c,t}^{\text{cur}}| - 1} \sum_{\boldsymbol{r}_{c,t} \in \mathcal{R}_{c,t}^{\text{cur}}} \left(\boldsymbol{r}_{c,t} - \boldsymbol{\mu}_{c,t}^{\text{cur}}\right)\left(\boldsymbol{r}_{c,t} - \boldsymbol{\mu}_{c,t}^{\text{cur}}\right)^{\top}. \tag{17}$$

The new covariance is updated online by the current mini-batch covariance and the old covariance

$$\boldsymbol{\Sigma}_{c,t}^{\text{new}} = \frac{(m-1) \cdot \boldsymbol{\Sigma}_{c,t}^{\text{old}} + (n-1) \cdot \boldsymbol{\Sigma}_{c,t}^{\text{cur}} + \frac{m \cdot n}{m+n} \cdot \boldsymbol{\delta}\boldsymbol{\delta}^{\top}}{m+n-1} \tag{18}$$

where m denotes the number of samples existing within the tth subclass of category c, $n = |\mathcal{R}_{c,t}^{\text{cur}}|$ indicates the number of samples in the current mini-batch, and $\boldsymbol{\delta} = \boldsymbol{\mu}_{c,t}^{\text{cur}} - \boldsymbol{\mu}_{c,t}^{\text{old}}$. When an epoch ends, the covariance corresponding to each prototype is replaced by its new one.

IV. Experiments

This section first outlines the experimental setting. Then, we provide a brief description of the comparative methods and evaluation metrics. Next, the three public CD datasets used are subsequently profiled. Finally, we present quantitative results and qualitative visualizations.

A. Experimental Setup

In general, when conducting semi-supervised CD experiments, our primary interest is on how the CD performance is impacted by different semi-supervised algorithms, rather than the improvement of network structure. Our segmentation network employs resnet50 as the backbone and pyramid pooling module (PPM) as the decoder. Note that we employ a dual-stream schema (Siamese network) to implement our method. To present a fair comparison between other state-of-the-art approaches and our proposed method, official default structures and parameter settings of each respective approach are kept unchanged.

We implement the proposed approach with PyTorch framework and conduct training/inference using a single NVIDIA GeForce RTX 3090Ti GPU with 24 GB. We optimize the parameters of the network by applying stochastic gradient descent (SGD) optimizer whose weight decay equals 0.0001 and momentum is 0.9. The initial learning rate is set as 0.001, and a polyschedule is employed to gradually adjust the learning rate to 0 until trained for 80 epochs. In the semi-supervised experiments, the value of batch size is fixed as 16 with eight labeled bitemporal image pairs and eight unlabeled bitemporal image pairs. The coefficient of unsupervised loss λ_1 is 1 and another of FDL λ_2 is 0.5 in (4). When it comes to dynamic updating of prototypes in (16), α is set as 0.999. Note that the threshold γ will be discussed in the ablation experiments.

To enhance the diversity of input data and facilitate consistency training in semi-supervised CD, some popular data perturbations are introduced. For labeled data, we perform weak perturbations such as image random flipping, image random cropping, and image random rescaling. Simultaneously, we take random blurring, color jitter, Gaussian blur, and grayscale as strong perturbations on unlabeled data.

B. Comparative Methods and Evaluation Metrics

In our experiments, the proposed novel approach and other recent supervised and semi-supervised methods are compared, among which there are fully supervised CD approaches: DMINet [43] and CSViG [10]. There are semi-supervised semantic segmentation approaches: s4GAN [44], FixMatch [45], CPS [37], ST++ [18], UniMatch [19], and CCVC [36]. There are recent semi-supervised CD approaches: RCR [25], RCL [20], and FPA [21].

To evaluate the CD performance of different semi-supervised algorithms, following another CD method [10] five evaluation metrics are presented in our experiment to establish qualities of different methods: intersection over union (IoU), $F1$ score, precision (Pre), recall (Rec), and overall accuracy (OA), where IoU and $F1$ score are the primary metrics, while the others are auxiliary metrics. To compare these semi-supervised methods, 5%, 10%, 20%, and 40% of data are randomly selected from the training set as labeled data, and the rest data are considered as unlabeled data to train the semi-supervised model.

C. Datasets

Three public CD datasets are used to assess the effectiveness of our method, which are LEVIR-CD [40], CDD-CD [41], and WHU-CD [42]. A comparative analysis of these datasets is shown in Table I. Note that all image pairs are high-resolution and captured in the visible light RGB mode.

1) LEVIR-CD: The LEVIR-CD [40], a public dataset focusing on changed buildings, consists of 637 bitemporal RS images, each measuring 1024 × 1024 pixels. These image pairs exhibit a spatial resolution of 0.5 m/pixel, capturing various changed building types, including residences, warehouses, apartments, and small garages. To facilitate fair comparisons with other CD approaches, we uniformly crop all original image pairs following a predefined split (7:1:2) into nonoverlapping patches with dimensions of 256 × 256. Consequently, we obtain 7120/1024/2048 pairs for training/validation/test, respectively.

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TABLE I
OVERVIEW INFORMATION ABOUT THE THREE PUBLIC CD DATASETS

Datasets	Image Type	Image Pairs	Image Size	Space Resolution	Train / Val / Test	Changing-objects
LEVIR-CD [40]	Satellite images	637	1024×1024	0.5 m/pixel	7120 / 1024 / 2048	building
CDD-CD [41]	Satellite images	16000	256×256	0.03 - 1 m/pixel	10000 / 3000 / 3000	building, forest, road ...
WHU-CD [42]	Aerial images	1	32207×15354	0.2 m/pixel	5947 / 743 / 744	building

TABLE II
AVERAGE QUANTITATIVE RESULTS (%) ON LEVIR-CD TEST SET WHEN DIFFERENT PERCENTAGES OF LABELED DATA ARE USED TO TRAIN DIFFERENT COMPARATIVE CD APPROACHES

Methods	5%					10%					20%					40%				
	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
DMINet [43]	86.15	72.39	78.67	64.84	98.35	88.54	81.35	84.79	73.60	98.78	89.21	83.87	86.50	76.15	98.90	**91.23**	86.94	89.03	80.23	99.10
CSViG [10]	88.38	71.98	79.34	65.76	98.43	89.61	74.61	81.42	68.67	98.57	89.82	83.03	86.29	75.88	98.89	90.79	86.83	88.76	79.80	99.08
s4GAN [44]	69.18	70.10	69.64	53.42	97.43	73.12	75.20	74.14	58.91	97.80	81.59	69.70	75.18	60.23	98.07	81.41	71.69	76.24	61.61	98.13
FixMatch [45]	86.76	81.79	84.21	72.72	98.71	88.66	83.49	85.99	75.43	98.86	90.45	83.99	87.10	77.15	98.96	89.91	84.12	86.92	76.87	98.94
CPS [37]	90.23	67.71	77.36	63.08	98.34	89.77	79.84	84.51	73.18	98.77	88.13	82.97	85.47	74.63	98.82	88.46	85.40	86.90	76.84	98.92
ST++ [18]	89.69	78.16	83.53	71.72	98.71	89.08	82.20	85.50	74.67	98.83	88.06	84.41	86.20	75.74	98.87	87.02	86.18	86.60	76.36	98.88
UniMatch [19]	89.84	82.13	85.81	75.15	98.86	89.47	85.10	87.23	77.35	98.95	89.52	85.65	87.54	77.84	98.98	90.15	85.60	87.82	78.28	99.00
CCVC [36]	**94.49**	56.59	70.78	54.78	98.04	90.49	83.21	86.70	76.52	98.93	**91.17**	84.31	87.61	77.95	99.00	89.95	85.53	87.69	78.08	98.99
RCR [25]	89.16	81.57	85.20	74.21	98.81	**90.58**	82.82	86.53	76.25	98.92	90.44	85.13	87.71	78.10	99.90	89.68	86.09	87.85	78.33	99.00
RCL [20]	87.55	83.63	85.54	74.74	98.81	86.92	86.81	86.86	76.78	98.90	86.16	87.76	86.95	76.92	98.89	85.45	88.85	87.12	77.17	98.90
FPA [21]	89.18	80.98	84.88	73.74	98.79	90.10	83.39	86.61	76.39	98.92	88.53	85.30	86.88	76.81	98.92	89.13	85.50	87.28	77.43	98.95
Sup. only	77.98	74.69	76.30	61.68	98.05	85.92	76.98	81.21	68.36	98.50	86.57	81.74	84.08	72.54	98.70	86.28	85.47	85.87	75.24	99.82
Ours	86.70	**88.36**	**87.52**	**77.81**	**98.94**	89.91	**87.01**	**88.44**	**79.27**	**99.05**	87.28	**91.25**	**89.22**	**80.54**	**99.07**	87.98	**90.93**	**89.43**	**80.88**	**99.10**

2) CDD-CD: The CDD-CD [41] stands as a public large-scale and challenging CD dataset curated from Google Earth. Comprising 11 pairs of high-resolution bitemporal satellite images, the spatial resolution ranges from 0.03 to 1 m/pixel, capturing a diverse array of changed areas, including man-made and natural changes, such as buildings, roads, cars, trees, and forests. In alignment with other datasets, we uniformly crop these image pairs into 256 × 256 pixel patches following the official division. Overall, the dataset consists of 16 000 small image pairs, distributed for training (10 000), validation (3000), and test (3000) samples.

3) WHU-CD: The WHU-CD [42] primarily focuses on the evolution of reconstructed buildings within the same area following an earthquake over several years. This dataset comprises a pair of aerial images sized 32 507 × 15 354 pixels, with a spatial resolution of 0.2 m/pixel, captured in 2012 and 2016. We followed the practice of cropping this image pair at the same scales, resulting in 7434 small image patches sized 256 × 256. During the experiment, we partitioned all data into 5947/743/744 pairs for training/validation/test, respectively.

D. Comparison With State-of-the-Art Methods

Subsequently, the experimental results of our proposed approach and other state-of-the-art semi-supervised CD methods on three publicly available datasets are analyzed in detail.

1) Quantitative Experimental Results on LEVIR-CD: The overall comparative results of the test set on LEVIR-CD [40] when the number of labeled images is increased from 5% to 40% are presented in Table II, where "Sup. only" means our model is trained with just the labeled data, and no unsupervised loss is calculated. From the table, we can observe that the performance of the fully supervised trained CD model is terrible, when the labeled data ares 5%, and $F1$/IoU/Pre/Rec/OA score is only 76.30%/61.68%/77.98%/74.69%/98.05%. With growing the number of labeled data, the improvement of CD performance in only fully supervised manner is particularly pronounced, IoU score reaches 61.68/68.36/72.54/75.24 when 5%/10%/20%/40% of the labeled data volume is utilized, respectively. This result also confirms that deep learning supervised algorithms rely heavily on massive training samples, and performance drops significantly when there are insufficient labeled data. However, semi-supervised methods can effectively alleviate this issue by using large amounts of unlabeled data. As depicted in Table II, these semi-supervised methods could accomplish better performance than the fully supervised manner in the same situation, and their CD performance further improves as the labeled data slowly increases. Our proposed method achieves the best CD performance in terms of $F1$, IoU, Rec, and OA metrics in the four sets of experiments, where the IoU score is above 2% higher than the recent FPA [21].

2) Quantitative Experimental Results on CDD-CD: Quantitative results implemented on CDD-CD [41] dataset are displayed in Table III. We first train a single student model on already selected labeled data as the baseline in a fully supervised manner. Then, other semi-supervised CD methods are performed by combining labeled and unlabeled data. As can be observed from this table, the proposed semi-supervised method achieves the highest score in both main evaluation metrics $F1$ and IoU. When 5% labeled data are served as the training set, $F1$/IoU/OA score of our semi-supervised method outperforms recent FPA by 2.07%/2.87%/2.86%/0.46%, and exceeds RCL [20] by 2.64%/3.64%/2.86%/0.88%, while $F1$/IoU/Pre/Rec/OA score is only 72.45%/56.80%/79.35%/66.65%/92.89% without utilizing unlabeled data. As the ratio of labeled data grows, the CD performance of all methods improves. Compared with a fully supervised baseline in the same condition using 40% labeled image pairs, the primary $F1$ and IoU scores of recent UniMatch are higher by 0.45% and 0.7%, RCR increases by 0.77% and 1.21%, and our proposed method improves by 3.71% and 5.97%.

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TABLE III

AVERAGE QUANTITATIVE RESULTS (%) ON CDD-CD TEST SET WHEN DIFFERENT PERCENTAGES OF LABELED DATA ARE USED TO TRAIN DIFFERENT COMPARATIVE CD APPROACHES

Methods	5%					10%					20%					40%				
	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
DMINet [43]	83.92	63.54	72.32	56.65	94.96	86.61	73.04	79.25	65.63	96.03	91.07	82.34	86.48	76.19	**97.33**	92.71	87.73	90.15	82.07	**98.01**
CSViG [10]	81.85	67.94	74.25	59.05	93.39	86.40	73.52	79.44	65.90	94.66	90.33	82.26	86.11	75.61	96.28	92.04	88.38	90.17	82.11	97.30
s4GAN [44]	78.80	58.35	67.05	50.43	91.95	83.26	66.97	74.23	59.02	93.48	87.27	69.59	77.43	63.18	94.31	86.08	70.51	77.52	63.30	94.26
FixMatch [45]	77.31	69.60	73.25	57.79	92.87	87.93	75.03	80.97	68.03	95.05	90.71	80.00	85.02	73.94	96.04	91.13	82.71	86.71	76.54	96.44
CPS [37]	**93.09**	46.85	62.33	45.28	92.06	92.60	61.24	73.72	58.38	93.88	91.94	68.84	78.73	64.92	94.78	**92.51**	79.32	85.41	74.53	96.20
ST++ [18]	84.59	71.72	77.63	63.43	94.20	88.98	76.46	82.24	69.84	95.37	88.89	82.05	85.33	74.41	96.04	89.32	84.51	86.85	76.76	96.41
UniMatch [19]	86.67	70.43	77.71	63.54	94.33	**91.10**	75.54	82.60	70.35	95.53	91.71	80.83	85.83	75.33	96.29	91.59	83.13	87.16	77.24	96.56
CCVC [36]	75.34	33.74	46.60	30.38	89.16	86.96	62.79	72.92	57.39	93.46	**93.97**	71.57	81.26	68.43	95.37	91.41	79.72	85.16	74.16	96.10
RCR [25]	86.16	71.06	77.88	63.78	94.34	88.63	77.99	82.97	70.90	95.51	90.51	83.20	86.70	76.52	96.42	90.66	84.52	87.48	77.75	96.61
RCL [20]	82.99	**74.20**	78.35	64.41	94.25	86.03	80.25	83.04	71.00	95.40	87.24	84.30	85.75	75.05	96.07	88.12	86.75	87.43	77.67	96.50
FPA [21]	88.65	71.12	78.92	65.18	94.67	89.75	76.65	82.68	70.48	95.50	90.42	81.53	85.74	75.05	96.20	90.33	83.06	86.54	76.27	96.38
Sup. only	79.35	66.65	72.45	56.80	92.89	85.43	71.23	77.69	63.52	94.26	87.26	78.92	82.88	70.76	95.43	89.39	84.19	86.71	76.54	96.38
Ours	89.46	73.98	**80.99**	**68.05**	**95.13**	90.50	**80.31**	**85.10**	**74.07**	**96.06**	91.04	**87.12**	**89.04**	**80.24**	96.99	91.40	**89.46**	**90.42**	**82.51**	97.34

TABLE IV

AVERAGE QUANTITATIVE RESULTS (%) ON WHU-CD TEST SET WHEN DIFFERENT PERCENTAGES OF LABELED DATA ARE USED TO TRAIN DIFFERENT SEMI-SUPERVISED APPROACHES

Methods	5%					10%					20%					40%				
	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
s4GAN [44]	84.93	76.70	80.61	67.52	92.99	82.66	80.26	81.44	68.69	93.05	86.60	81.54	83.99	72.40	94.10	87.59	81.52	84.44	73.07	94.29
FixMatch [45]	90.84	71.00	79.70	66.25	93.13	94.12	81.12	87.14	77.21	95.45	90.84	87.53	89.15	80.43	95.95	95.07	84.74	89.61	81.18	96.27
CPS [37]	91.34	76.26	83.12	71.11	94.12	91.04	82.48	86.55	76.29	95.13	92.51	85.85	89.05	80.27	95.99	94.23	85.46	89.63	81.21	96.24
ST++ [18]	90.61	83.17	86.73	76.58	95.17	90.84	85.96	88.34	79.11	95.69	92.13	89.30	90.70	82.98	96.52	91.18	91.03	91.11	83.66	96.62
UniMatch [19]	**93.96**	78.03	85.26	74.30	94.87	91.91	84.03	87.79	78.24	95.56	**93.99**	86.34	90.00	81.82	96.36	94.86	86.20	90.32	82.35	96.49
CCVC [36]	91.28	79.94	85.23	74.27	94.74	92.03	82.14	86.80	76.68	95.26	92.83	84.82	88.64	79.60	95.87	93.28	87.04	90.05	81.91	96.35
RCR [25]	91.62	83.27	87.25	77.38	95.38	89.13	89.19	89.16	80.44	95.88	91.90	90.19	91.04	83.55	96.63	90.70	92.43	91.56	84.43	96.76
RCL [20]	90.95	86.26	88.54	79.44	95.76	91.49	90.13	90.80	83.16	96.53	91.79	91.56	91.67	84.63	96.84	91.80	91.90	91.85	84.93	96.90
FPA [21]	90.54	87.87	89.19	80.48	95.95	91.05	89.32	90.18	82.11	96.30	92.97	90.24	91.59	84.48	96.85	91.16	92.75	91.95	85.10	96.92
Sup. only	81.67	76.69	79.10	65.43	92.30	83.36	85.57	84.45	73.09	94.02	85.36	88.44	86.87	76.79	94.92	89.01	90.18	89.59	81.14	96.02
Ours	92.77	**89.57**	**91.14**	**83.72**	**96.69**	**94.79**	**90.92**	**92.82**	**86.60**	**97.33**	93.89	**93.40**	**93.65**	**88.05**	**97.59**	**95.51**	**93.25**	**94.37**	**89.34**	**97.89**

3) Quantitative Experimental Results on WHU-CD: We keep in line with the RCL [20] approach for the WHU-CD [42] dataset, removing samples without any changes and then conducting experiments in the subset of WHU-CD to facilitate better comparisons. Table IV presents the overall experimental results when the number of labeled images increases from 5% to 40%. The performance of all methods demonstrates a clear upward trend as the number of labeled samples increases. Notably, the fully supervised method shows a significant enhancement, where the IoU score increases from 65.43% with 5% labeled data to 81.14% with 40% labeled data. Our approach consistently outperforms other methods in all conducted experiments, and we attribute this success to our innovative PDL. This method effectively enhances intraclass compactness and interclass variability, contributing to superior overall performance. When only 5% labeled images are used, the $F1$/IoU score exceeds RCL by 2.60/4.28% and outperforms FPA by 1.95%/3.24%, and the IoU score about change area is 18.29% higher compared with the fully supervised method.

4) Visualization Results: To facilitate an intuitive comparison of detection performance among different semi-supervised CD methods, we provide visualizations of qualitative prediction results on test images from three CD datasets. This comparison is conducted under the uniform condition of utilizing a 20% proportion of labeled data for each method. As presented in Fig. 4, for better discrimination and observation, different rendering colors are used to indicate true positives (white), true negatives (black), false positives (red), and false negatives (green). Semi-supervised semantic segmentation models trained with small volumes of labeled data lead to numerous false positives and false negatives, as well as ambiguous information about the boundaries of changed objects becomes more apparent. However, these qualitative results indicate that semi-supervised CD methods can achieve more accurate predictions. Moreover, our proposed approach exhibits higher prediction accuracy and less error than other semi-supervised CD techniques. For example, other comparative methods fail to completely detect the complex changes of roads in Fig. 4(d), but our proposed method not only effectively detects the changed roads but also ensures better connectivity between them. Furthermore, in (f), s4GAN [44], FixMatch [45] and CPS [37] exhibit some dismal false negatives, conversely CCVC [36], RCR [25], RCL [20], and FPA [21] detect unexpected false positives, and our method can precisely and entirely identify real changed areas.

In addition, we select a pair of bitemporal test images from each used dataset, then present their qualitative visualizations when different semi-supervised CD methods are trained using various proportions of labeled data in Fig. 5(a)–(k) denote results of different CD approaches for three examples at 5%, 10%, 20%, and 40% labeled data, respectively. From this figure, the amount of labeled data can significantly affect CD performance, which illustrates that deep learning methods rely on the volume of training samples. With the same number of labeled and unlabeled images used to train models, our method exhibits more accurate predictions in the boundary of changed areas than other recent semi-supervised CD approaches. Therefore, both quantitative experimental

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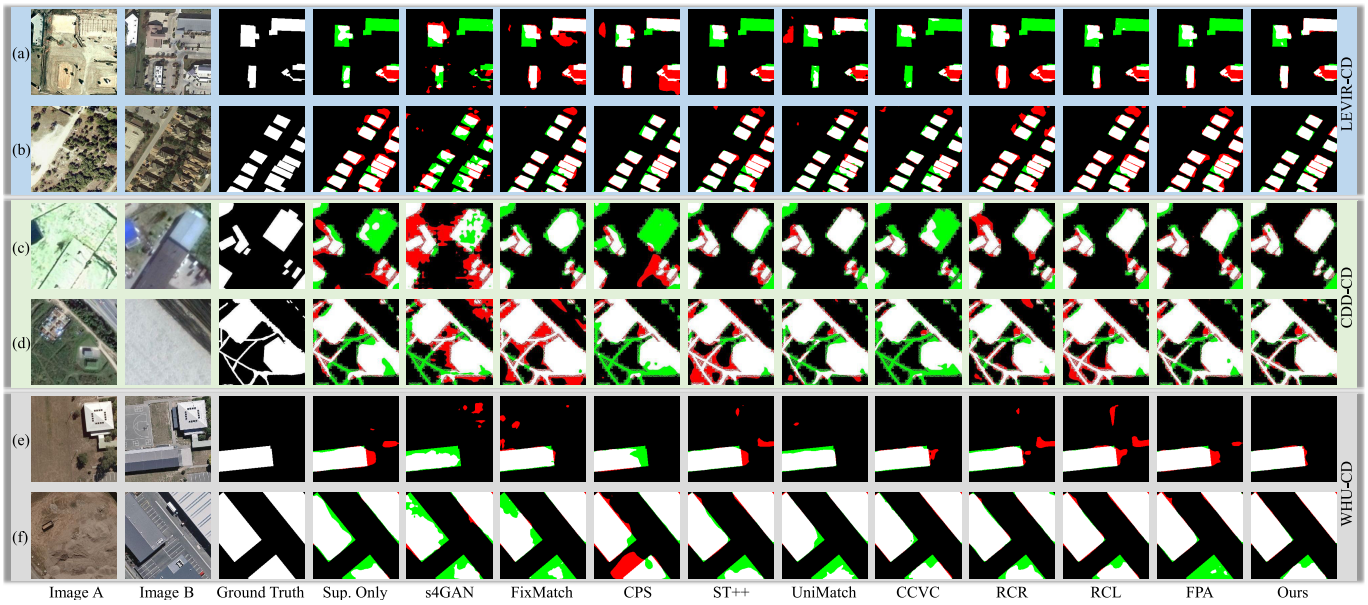


Fig. 4. Qualitative visualizations of different CD methods on three datasets. The presented colors indicate true positives (white), true negatives (black), false positives (red), and false negatives (green). (a)–(f) Different test examples.

data in the previous subsections and qualitative visualization Sections IV-D1–IV-D3 sufficiently illustrate that our proposed semi-supervised CD method presents superior performance in the changed areas of interest.

V. Discussion

In this section, we provide more experimental results on the CDD-CD dataset to further discuss and analyze our proposed approach. First, ablation experiments are performed to demonstrate the effectiveness of PDL, followed by a discussion on the influence of prototype numbers on PDL and an analysis of threshold settings for reliable pseudolabel selection. Next, we compare the generalization of different CD methods. Furthermore, the efficiency of comparative semi-supervised CD methods in parametric number and time complexity is compared. More detailed descriptions are given below.

A. Ablation Study

1) Effectiveness of Prototype Discriminative Learning: Based on the ST [16] as a basic semi-supervised framework, we propose PDL in the deep feature domain to improve the compactness of pseudolabel within each category and the variability of different categories, and we also conduct comparison experiments to verify its effectiveness. All experimental results are reported on the CDD-CD dataset using 5% labeled data as depicted in Table V. For (a), the model is trained with a full supervision manner using only a small fraction of labeled data. Subsequently, ST combines massive unlabeled data and the 5% labeled data together to train the semi-supervised model as a baseline, $F1$/IoU/OA score is 79.79%/66.38%/94.81% as shown in (b). Next, we incorporate PCL into the baseline to train the semi-supervised model, and $F1$/IoU/OA score is improved by 1.20%/1.67%/0.32%. In addition, CutMix as the most commonly used data augmentation in semi-supervised segmentation tasks is also discussed.

TABLE V
Effectiveness of the Different Components When Model Is Trained on CDD-CD Dataset Using 5% Labeled Data

Methods	Sup	ST	PDL	CutMix	F1 (%)	IoU (%)	OA (%)
(a)	✓				72.45	56.80	92.89
(b)	✓	✓			79.79	66.38	94.81
(c)	✓	✓	✓		80.99	68.05	95.13
(d)	✓	✓		✓	80.78	67.76	95.13
(e)	✓	✓	✓	✓	**81.63**	**68.91**	**95.32**

In (d), there is a significant performance improvement when employing the baseline with CutMix for training. Particularly, when PDL and CutMix are used together, $F1$/IoU/OA score exceeds (b) by 1.84%/2.53%/0.51%, and outperforms (c) by 0.64%/0.86%/0.19%. To further verify the effectiveness of the proposed PDL, we present the visualization of feature embedding in Fig. 6. It can be observed that our method can promote the compactness of intraclass feature distribution and ensure variability of interclass features. This is mostly attributed to the use of PDL. Therefore, both quantitative experimental data and qualitative visualization validate its effectiveness.

In the ST semi-supervised learning framework, the pseudolabel generated by the teacher model guides the student model to fully exploit useful information from the unlabeled data as a supplement. When plenty of unreliable pseudolabels are applied, student model might learn incorrect information leading to a collapse. Therefore, the performance of semi-supervised methods is mirrored in the quality of generated pseudolabel to a certain extent. Fig. 7 illustrates the comparative results of pseudolabel quality between baseline and our approach on CDD-CD. The analysis from this figure reveals that the pseudolabel generated by our proposed method demonstrates increased reliability compared with the baseline, which is attributed to the fact that PDL is used. In addition,

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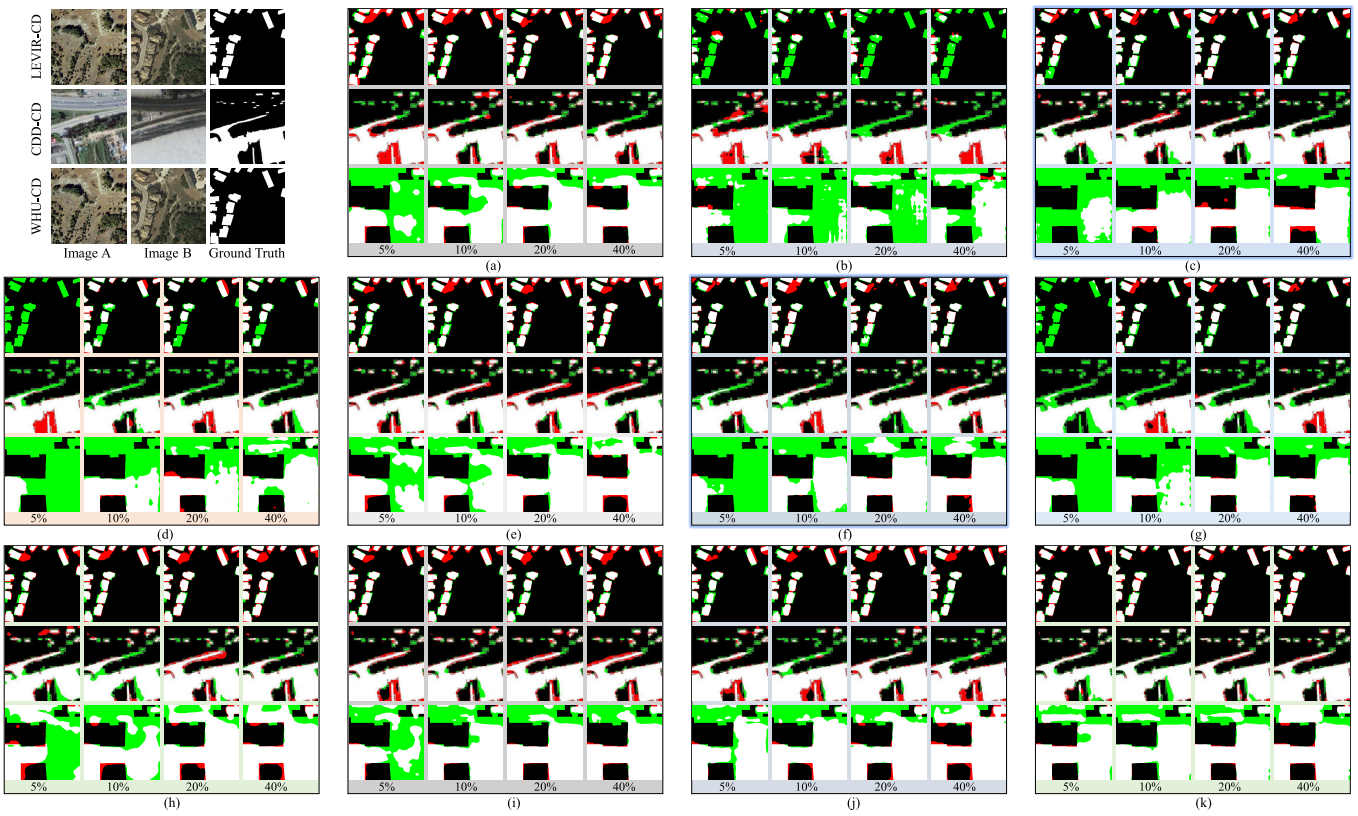


Fig. 5. Qualitative visualizations of some examples obtained from different methods using various percentages of labeled data trained on three datasets. (a) Sup. Only. (b) s4GAN. (c) FixMatch. (d) CPS. (e) ST++. (f) UniMatch. (g) CCVC. (h) RCR. (i) RCL. (j) FPA. (k) Ours.

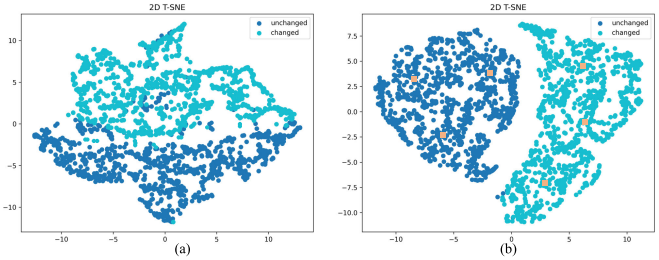


Fig. 6. Feature embedding visualizations trained on the 5% labeled data of CDD-CD dataset using T-SNE. (a) Sup. Only. (b) Ours (■ denotes prototypes).

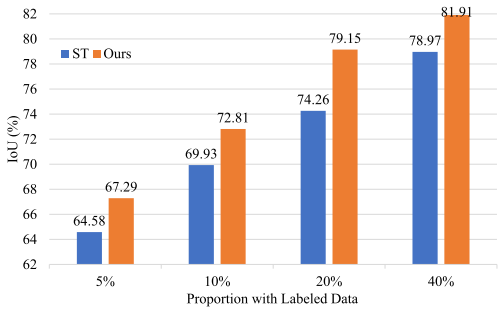


Fig. 7. Quantitative results of pseudolabel generated by teacher model when our proposed method and baseline are trained on CDD-CD with different proportions of labeled data.

we also compare the qualitative visualization of generating pseudolabel for unlabeled data in multiple epoch stages as shown in Fig. 8. The pseudolabel generated by our teacher model with PDL is significantly more reliable than the native ST semi-supervised framework. Therefore, these results strongly support the effectiveness of our PDL.

2) Influence of the Number of Prototypes: In our proposed approach, each category is divided into multiple subclasses in the deep feature domain, and we perform experiments to explore the influence of the number of prototypes on CD performance. We use 5% labeled data on CDD-CD for analysis by two groups of experiments. As shown in Table VI (NP^u and NP^c represent the number of prototypes in changed and unchanged categories), we first configure the number of prototypes in changed and unchanged categories to be the same. When each category has only one prototype in the deep feature domain, the $F1$/IoU/OA score is 80.04%/66.72%/94.92%,

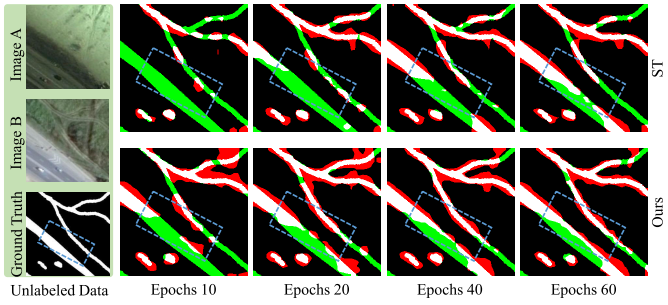


Fig. 8. Qualitative visualization of pseudolabel generated by teacher model for an unlabeled sample in different epochs when our method and baseline are trained using 5% labeled data on CDD-CD dataset.

which is 0.25%/0.34%/0.03% higher than baseline without PDL. With an increasing number of the same prototypes, the

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TABLE VI

COMPARISON OF MODEL PERFORMANCE WHEN DIFFERENT NUMBERS OF PROTOTYPES ARE SET FOR EACH CATEGORY

Methods	NP^c	NP^u	F1 (%)	IoU (%)	OA (%)
Baseline	-	-	79.79	66.38	94.81
Same Num.	1	1	80.04	66.72	94.92
	3	3	80.99	68.05	95.13
	5	5	81.24	68.41	95.16
	7	7	79.49	65.96	94.73
Diff. Num.	1	2	81.03	68.11	95.10
	1	3	80.75	67.72	95.09
	2	4	80.25	67.01	94.91
	3	5	80.03	66.70	94.91

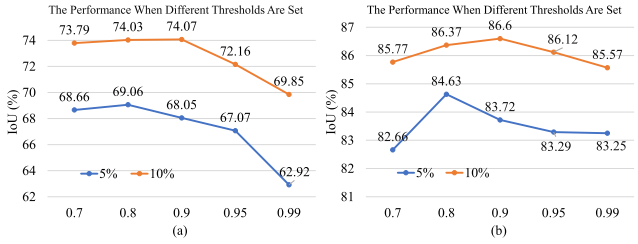


Fig. 9. Experimental results using 5% and 10% labeled data on CDD-CD and WHU-CD datasets when different γ values are set, respectively. (a) CDD-CD. (b) WHU-CD.

CD performance of our model becomes better, reaching a peak at the number of prototypes as 5, after which CD performance begins to drop when the number is further increased. When the changed and unchanged categories possess different numbers of prototypes, a slight fluctuation in performance can be observed, but they are always ahead of the baseline. It should be stated that we chose a relatively simple way, employing the same number of prototypes as 3 for each category in all other experiments of three datasets.

3) Threshold for Selecting Reliable Pseudolabel: Although the teacher model in our semi-supervised method is able to generate more reliable pseudolabels for unlabeled data, the use of all pseudolabels containing incorrect information misleads the trained model toward an opposite goal. Therefore, we set the threshold γ to select pseudolabels with appropriate quantity and quality. In general, selecting a low threshold would involve too many pseudolabeled data in the model training process, resulting in overfitting to mislabeled data. Conversely, a high threshold might restrict the inclusion of pseudolabeled data, causing underfitting during model training. Fig. 9(a) and (b) presents the experimental results of our method on CDD-CD and WHU-CD datasets using 5% and 10% labeled data when different values of γ are set, respectively. In (a), for the 5% labeled images, thresholds of 0.8 and 0.7 accomplish optimal and suboptimal results, respectively; 0.9 and 0.8 become the top two scores at 10% of labeled data, respectively. When it comes to the WHU-CD in (b), the optimal thresholds are 0.8 and 0.9 when using 5% and 10% labeled data, respectively. Consequently, it should be noted that we adopt by default a relatively moderate threshold of 0.9 for other experiments on all used datasets.

B. Generalization of Methods

To verify the generalization of our proposed semi-supervised method, we compare IoU scores of different methods when training and test sets come from different datasets, and the quantitative data are presented in Table VII. First, the model is trained using a training set of CDD-CD and tested on the test set of the WHU-CD dataset. It should be noted that changed objects included in these two datasets are quite different, CCD-CD contains man-made and naturally changed objects of various sizes, while CDD-CD mainly covers some rebuilt man-made buildings. Therefore, it can be seen from this table that the performance of all semi-supervised methods in this group of experiments is poor in general, but the performance of our method is consistently higher than other recent methods. Next, the training set of LEVIR-CD and the test set of WHU-CD are used for further comparison. Because both datasets only contain changed buildings, the experimental results for this group are more substantial than the previous group, and the performance of our method is optimal except for using 40% labeled data. Thus, the experimental results in Table VII verify that our model has better generalization ability compared to other recent semi-supervised CD methods.

TABLE VII

IOU(%) OF DIFFERENT CD METHODS ABOUT GENERALIZATION WHEN DIFFERENT PROPORTIONS OF LABELED DATA ARE USED

Methods	$CDD\text{-}CD_{(train)} \rightarrow WHU\text{-}CD_{(test)}$				$LEVIR\text{-}CD_{(train)} \rightarrow WHU\text{-}CD_{(test)}$			
	5%	10%	20%	40%	5%	10%	20%	40%
DMINet [43]	16.35	15.33	17.86	18.36	35.89	41.56	34.66	51.71
CSViG [10]	9.25	9.68	11.25	12.58	33.00	37.30	36.15	46.55
s4GAN [44]	13.76	15.47	14.92	13.55	22.50	19.22	27.85	28.01
FixMatch [45]	18.05	17.66	20.54	18.26	31.99	38.81	41.67	46.37
CPS [37]	9.03	15.02	9.37	13.90	15.10	22.30	16.97	20.71
ST++ [18]	18.80	21.07	21.90	19.71	17.18	19.77	40.40	41.01
UniMatch [19]	18.33	17.35	21.28	20.40	30.39	40.62	41.40	40.86
CCVC [36]	5.50	16.15	15.03	12.65	13.19	31.89	36.54	31.42
RCR [25]	16.74	19.16	19.38	18.64	37.29	40.38	45.93	46.83
RCL [20]	17.79	20.31	21.39	20.48	32.00	43.15	40.76	51.98
FPA [21]	16.12	17.98	17.45	16.93	37.51	41.72	44.84	**52.65**
Ours	**19.66**	**21.99**	**22.47**	**20.62**	**39.45**	**45.32**	**47.55**	47.78

TABLE VIII

TIME COMPLEXITY ANALYSIS OF DIFFERENT CD METHODS USING 5% LABELED DATA ON CDD-CD

Methods	Params (M)	Train times (h)	F1 (%)	IoU (%)	OA (%)
DMINet [43]	6.24	0.73	72.32	56.65	94.96
CSViG [10]	37.96	2.75	74.25	59.05	93.39
s4GAN [44]	26.54	6.35	67.05	50.43	91.95
FixMatch [45]	40.47	10.92	73.25	57.79	92.87
CPS [37]	80.95	14.38	62.33	45.28	92.06
ST++ [18]	46.85	6.60	77.63	63.43	94.20
UniMatch [19]	40.47	15.27	77.71	63.54	94.33
CCVC [36]	80.95	14.13	46.60	30.38	89.16
RCR [25]	50.69	17.07	77.88	63.78	94.34
RCL [20]	46.85	14.42	78.35	64.41	94.25
FPA [21]	46.85	14.26	78.92	65.18	94.67
Ours	46.85	17.53	**80.99**	**68.05**	**95.13**

C. Time Complexity

Although the substantial enhancements achieved by our proposed approach in detecting changed areas with a limited labeled dataset and a substantial amount of unlabeled data, a detailed analysis of its parameters and complexity is essential. Table VIII presents the relevant data, since an extremely limited number of labeled samples are utilized, training time in a fully supervised manner is the least but there is substantial potential for improvement in the CD performance.

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The semi-supervised methods expand the training samples to explore supplementary information from the massive unlabeled data; thus, they usually consume more training time. Since RCR [25] uses consistency regularization to compute unsupervised losses between several auxiliary encoders and the main encoder, its number of parameters increases. Compared with RCR, training time for RCL [20] and FPA [21] is shortened by over 2 h, with only a slight increase in scores of evaluation metrics. Among these semi-supervised methods, our proposed approach avoids the introduction of extra parameters, and the training time experiences a slight increase, this is influenced by the division of prototypes within categories by the pretrained student model and continuous updating of the prototypes. However, our semi-supervised CD method attains the most superior performance. It should be noted that all semi-supervised CD methods exhibit identical inference times.

VI. Conclusion

In this work, we propose a new semi-supervised CD approach, which harnesses information from an extensive set of unlabeled RS data as a complement for improving the generalization and robustness of the model. To exploit invisible and helpful feature properties in the deep feature domain, FDL is utilized to build intraclass compactness and interclass variability in PDL, which divides each category into multiple subclasses whose feature distributions obey a Gaussian distribution, encouraging the features of samples closer to the nearest prototype while away from all prototypes of other categories at the deep feature level. Finally, a series of experiments are performed on three widely used datasets to prove the effectiveness of our method. The experimental data consistently confirm its commendable generalization capabilities and robust performance.

In future work, we plan to further explore unreliable pseudolabels to increase the variability between categories, besides utilizing reliable pseudolabels with high confidence. Although the semi-supervised CD algorithm alleviates the dependence of deep models on the amount of labeled training data at a certain level, a few labor-intensive pixel-level annotations are still required. Therefore, it is also a challenging task to explore weakly supervised or unsupervised CD frameworks by employing more economical coarse supervision signals or using the foundation models.

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